

Bellman-Ford Algorithm :- SSSP with +ve or -ve weights

$\text{Relax}(u, v, w) \rightarrow$ edge $u \xrightarrow{w} v$, w is weight of the edge

if $v.d > u.d + w(u, v)$ old weight of v . ($v.d$)

$v.d = u.d + w(u, v)$
 $v.\pi = u$

Compare with

$(u.d)$ weight to reach u +
weight of $u \rightarrow v$ edge
 $w(u, v)$

Bellman-Ford(G, s)

$s.d = 0$

for all other vertices $v.d = \infty$, $v \in G.V - s$
 $v.\pi = \text{Null}$

for $i = 1$ to $|G.V| - 1$

for each edge $u \rightarrow v$ in $G.E$

$\text{Relax}(u, v, w)$

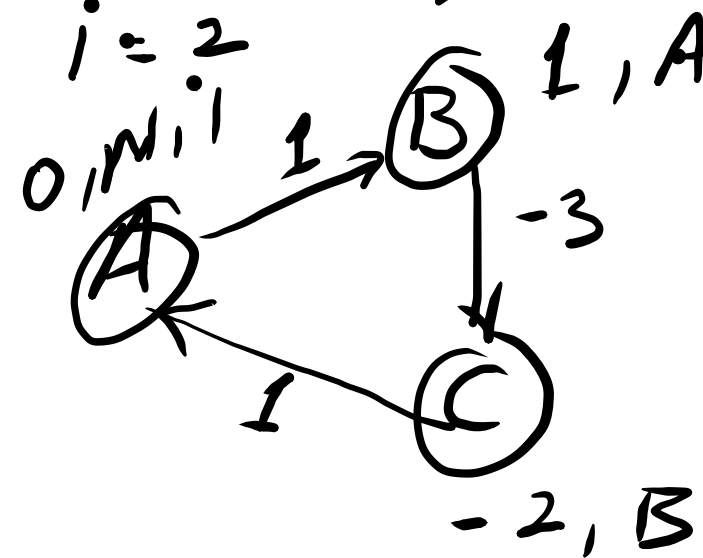
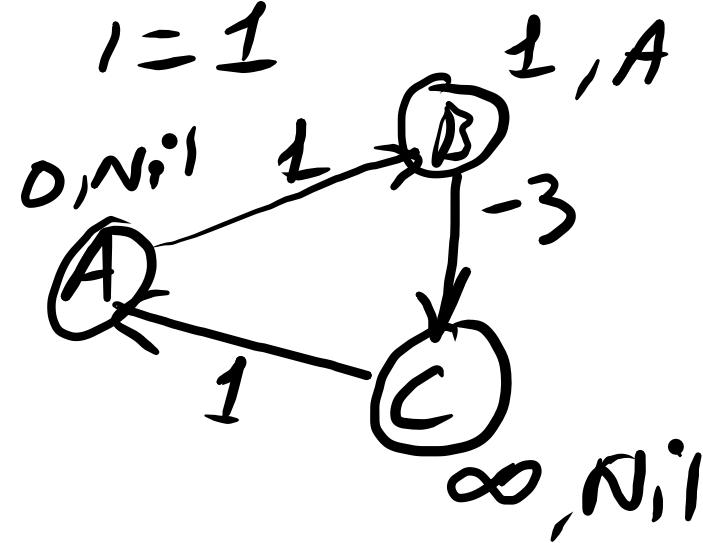
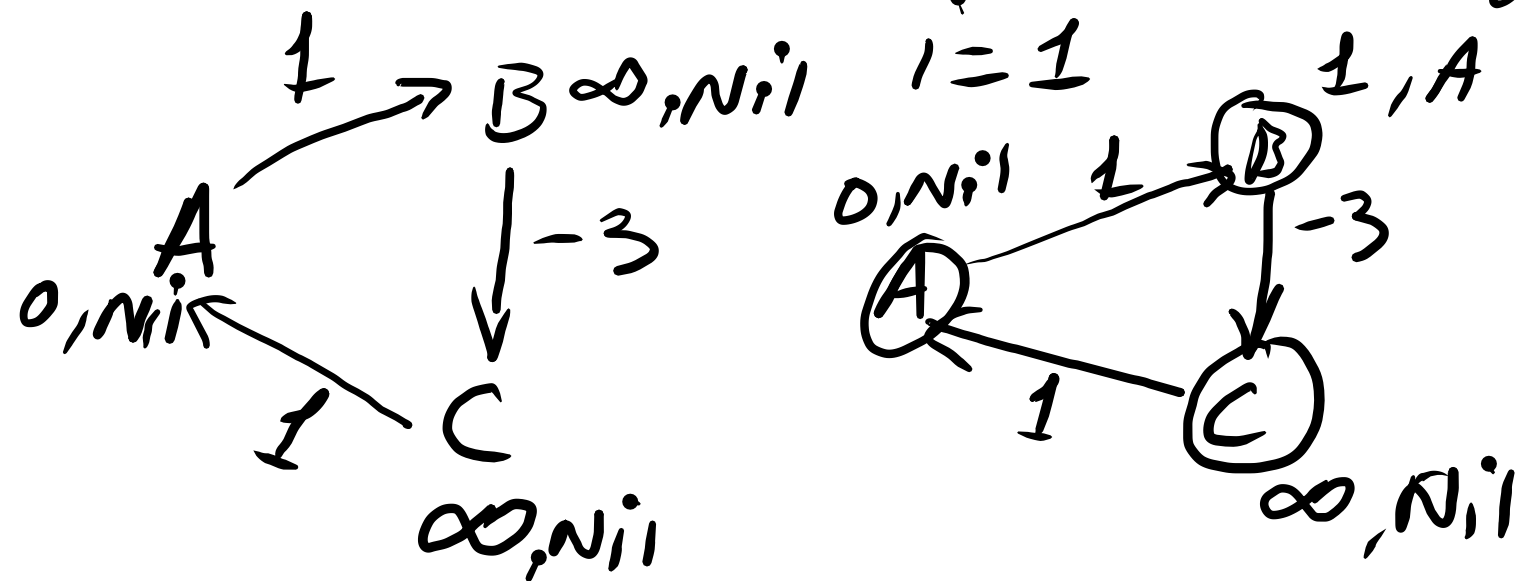
Bellman-Ford(G, s)

for $i = 1$ to $V - 1$

for each edge $u \rightarrow v$ in $G.E$
 $\text{Relax}(u, v, w)$

for each edge $u \rightarrow v$ in $G.E$
 if $v.d > u.d + w(u, v)$

error('Negative-weighted cycle found')



$C \rightarrow A$
 $A.d > C.d + w(C, A)$

$$0 > -2 + 1$$

$$-1$$

